

Passive Face Liveliness Detection

Umesh Gupta¹

SCSET, Bennett University
Greater Noida, India
er.umeshgupta@gmail.com
0000-0002-1547-7974

Kanav Arora²

SCSET, Bennett University
Greater Noida, India
kanavlt885@gmail.com
0009-0003-9374-2184

Sehaj K Khurana³

SCSET, Bennett University
Greater Noida, India
sehajkkhurana@gmail.com
0009-0005-3502-1666

Nilay Gupta⁴

SCSET, Bennett University
Greater Noida, India
nilaygupta3003@gmail.com
0009-0001-0125-6126

Nikunj Gupta⁵

SCSET, Bennett University
Greater Noida, India
nikunjgupta609@gmail.com
0009-0002-1317-263X

Abstract—Facial liveliness detection is an important component of today's authentication systems to identify live faces and spoof attempts, ensuring safe and accurate identification. Combating facial spoofing attacks in terms of printed photos, videos, or 3D masks has become necessary with digital interactions. This paper presents a comprehensive study on the development of a multimodal model for passive face liveliness detection using diverse datasets and Convolutional Neural Network(CNN) architectures. It utilizes multiple input modalities, including texture features, spatial information, and depth maps, in order to enhance its capability to detect spoofing attempts effectively. The system integrates multimodal data, which ensures robust performance and detection under changing conditions or diverse environments such as changes in illuminations, pose variations, occlusions, and so on. Furthermore, advanced preprocessing techniques comprise methods such as image normalization and augmentation, allowing sufficient preparation of the input for actual model training. The use of the well-established ResNet-50 architecture enables the model to use rich, discriminative features to identify fine and subtle differences between live faces and attempts. The model is tested extensively using publicly available datasets for performance evaluation. Comparisons with existing approaches, such as Transformer-based vision models, MobileNetV2, and DenseNet-121, show how the proposed approach works better and is more accurate and reliable. With great emphasis on real-world application, the study shows adaptability and precision of this model under diverse settings and tasks. The proposed system contributes toward safe and secure authentication technologies in providing high accuracy and robust detection performance. This work, hence, emphasizes the need for a multimodal approach and the use of advanced architectures in dealing with the increased face spoofing challenges in the security-centric digital world of today.

Index Terms—Passive face liveliness, Eigen-faces, Fisherface, Convolutional neural network, Vision transformer, MobilenetV2

I. INTRODUCTION

Currently, facial images are extensively employed as data across multiple sectors, including surveillance, office environments, and smartphone authentication [1], among others. The technique of face recognition has developed into one of the leading methods for secure access control, coinciding with

the increasing availability of biometric authentication systems. Nonetheless, the increasing implementation of facial recognition technology has raised significant concerns regarding its vulnerability to spoofing attacks, wherein individuals without authorization attempt to circumvent security mechanisms by mimicking legitimate users through images, videos, or three-dimensional models [2], [3]. To reduce these risks, face liveliness detection has emerged as an integral component of modern biometric systems to ensure that the face under examination belongs to a real, living person rather than a fake representation [4].

Face liveliness detection has traditionally relied on active methods, requiring users to perform actions like smiling, blinking, or head movements to confirm presence [5]. While effective, these methods can hinder user experience by demanding effort. Passive detection overcomes this by assessing subtle facial features, movements, and contextual cues to verify if the face is real [6].

The fundamental benefit of passive liveliness detection is the ability to provide continuous, non-intrusive protection while retaining a high level of accuracy. With advances in deep learning, computer vision, and sensor technologies, passive facial liveliness detection has received a lot of attention in recent years [7], [8], opening up new possibilities for improving security as well as accessibility of biometric authentication systems. This study investigates the present methodologies, problems, and future possibilities in passive face liveliness detection, emphasizing its potential to transform the way face recognition is used in real-world applications [9]. The following are the contributions of the proposed work:

- **Innovative multimodal architecture development:** A core model architecture that incorporates CNN-based methods for facial liveliness detection has been designed, enabling precise and effective performance assessment. Architectural enhancements have also been explored to improve efficiency, accuracy, and adaptability across diverse use cases.
- **Comprehensive dataset creation and preprocessing:**

To improve model training and optimize performance through fine-tuning, a variety of datasets have been developed and data preprocessing techniques have been utilized.

Numerous limitations found in the research publications examined during the literature assessment emphasize the necessity of the proposed methodology. Several face feature extraction projects currently in existence rely on antiquated architectures or conventional machine learning methods, which fall short of utilizing the full potential of contemporary deep learning innovations such as CNNs [7], [11], [12]. Furthermore, the majority of studies used homogeneous or small datasets, which resulted in models that are not generalizable and have trouble handling a variety of real-world situations [8], [13]. Additionally, hyperparameter tweaking and optimization techniques were frequently disregarded, which led to scaling problems and less-than-ideal performance [15], [17]. These limitations highlight the need for our study, which presents a strong CNN-based model architecture that has been trained on a variety of excellent datasets. This research intends to improve model flexibility under various settings and increase the accuracy of facial feature extraction by tackling these restrictions.

This paper is organized to present the research in a clear and systematic manner. It begins with an introduction highlighting the significance of face liveness detection and its various types, followed by a review of related work conducted in this domain. The proposed method is then detailed, including an analysis of existing approaches, the integration of depth analysis, the adopted algorithm, and the underlying mathematical framework [14], [16]. The result and discussion section provides a comprehensive overview of the findings, comparing various viable methods for face liveness detection. Finally, the conclusion summarizes the paper and offers insights into potential directions for future work.

II. RELATED WORK

Face recognition has evolved from early methods like Eigenfaces and Fisherfaces, which relied on linear transformations for feature extraction, to modern deep learning-based techniques that achieve state-of-the-art performance [1], [7]. However, even with such advancements, face recognition systems remain susceptible to spoofing attacks, including print, video replay, and 3D mask attacks, which threaten their reliability in secure environments [3], [8].

To counter these vulnerabilities, liveness detection has become an essential component of biometric systems. Early approaches predominantly relied on active techniques, requiring users to perform specific actions, such as blinking, smiling, or nodding, to confirm liveness [5]. While effective, these methods introduced usability challenges by demanding explicit user interaction.

Modern works have increasingly focused on the use of passive liveness detection approaches that analyze the low-level facial cues, movement dynamics, and temporal variations for the distinction between real and spoofing faces without

any intervention of the user [6]. For example, Feng et al. [6] utilized movement pattern analysis, whereas Liu et al. [7] used deep tree learning for boosting detection performance in zero-shot scenarios. Qin et al. [8] also used invariant feature representations to improve the detection accuracy under different environmental conditions.

Multimodal information, including depth, thermal imagery, and contextual information, has been integrated into passive detection schemes, making them more robust and reliable. Modern research works, including that by Stehouwer et al. [10], employ advances in GAN-based approaches to simulate adversarial attacks and enhance the robustness of the detection system against advanced spoofing attacks.

Despite these advancements, there still exist challenges to the possibility of ensuring consistent performance in diverse conditions, on different devices, via varying attack types. This research investigates recent breakthroughs and deficiencies of practical relevance in passive liveness detection.

III. PROPOSED WORK

In order to achieve optimal results, the input data is utilized as depicted in Fig. 1. The same is presented in a stepwise manner in Table I.

TABLE I. ALGORITHMIC FLOW OF METHODOLOGY

Step	Description
Step 1	Input data is provided
Step 2	Detects faces using algorithms
Step 3	Preprocesses detected faces
Step 4	Performs texture analysis
Step 5	Analyzes depth features
Step 6	Splits video input into frames
Step 7	Extracts features with ResNet-50
Step 8	Trains the model using features
Step 9	Tests and validates the model
Step 10	Ends the process if successful

The process starts with Input Data consisting of raw images or video frames. Face detection helps to identify and locate the faces, which are then followed by preprocessing wherein the detected faces are normalized through resizing or filtering, among others. Texture analysis extracts surface features to distinguish between real and fake faces, while depth analysis detects depth cues to identify 3D structures. Video input is split into frames for further analysis. Feature extraction with ResNet-50 extracts deep learning-based features, which are utilized in the training of the model to distinguish between live faces and artificial ones. Following this stage, the model is subjected to a test and validation process to determine its performance. If the system finds a requirement for better performance, then further compilation of textural and depth results is carried out to improve the accuracy. When the improvement in performance is attained, success is achieved, and the pipeline is completed at this stage.

We have used the popularized convolutional neural network, ResNet-50, as a backbone for face liveness detection. ResNet-50 eradicates the vanishing gradient problem through residual blocks that make the deep networks train better than ever before [11]. With 50 layers, the model captures all the intricate

facial characteristics, for instance, reflectance, texture of skin, and patterns of skin which can help differentiate real faces from the spoofed ones, such as printed photos or videos [12]. Pre-trained on ImageNet, it serves as a good starting point for our work and enables robust generalization [13].

To make the system more resilient, we have incorporated depth analysis as another module. This module evaluates the 3D spatial structure of a face, differentiating live faces with natural depth fluctuations and flat 2D spoof artifacts [14]. The depth analysis pipeline accepts scaled input frames and uses lightweight convolutional layers in order to extract a depth score that quantifies the spatial characteristics of the face [15]. The depth analysis module is designed to ensure detection of structural inconsistencies that normally are disregarded when relying upon texture analysis [16].

We have developed a multimodal liveness detection framework by fusing the depth analysis module with the feature extraction capabilities of ResNet-50. While depth analysis detects 3D discrepancies [14], ResNet collects fine-grained textural details [11]. When combined, these elements provide stronger performance against complex spoofing tactics, fewer false positives, and increased detection accuracy [16]. This hybrid method considerably increases the dependability of the system in applications such as online payments, secure authentication, and eKYC procedures [12].

A. Algorithm methodology

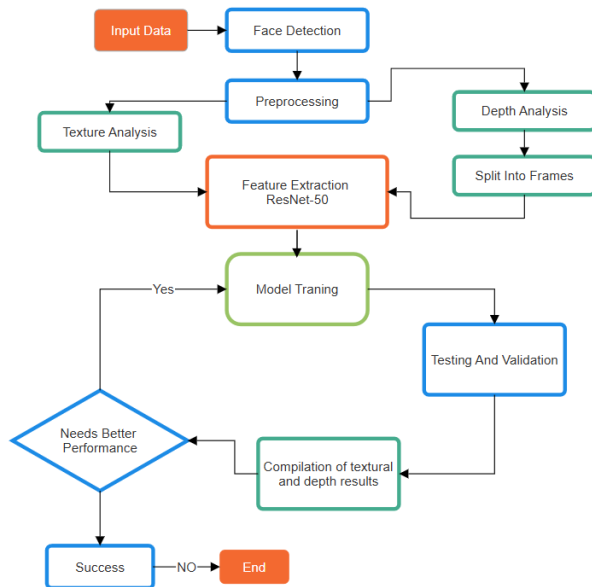


Fig. 1. Example of control flow for the proposed methodology

• Webcam image capture

The `colab_webcam` function enables taking a picture using a computer's built-in camera when working in a Google Colab notebook [17]:

- It uses **JavaScript** to access the webcam, stream the video feed, and capture a frame as a JPEG image.

- The photographed image is converted from its base64-encoded format into a NumPy array compatible with OpenCV for further processing.

• Depth analysis model

The `DepthAnalysisModel` is a placeholder neural network for depth analysis [14], [15]. It consists of:

- A **convolutional layer** followed by **ReLU activation** [16].
- A **fully connected layer** to predict depth from a resized face crop (assumed to be 64x64) [14].

• Local binary patterns (LBP) for Texture Analysis

The `texture_analysis` function uses **LBP** to analyze the texture of a face crop [18], [19]:

- Converts the face to **grayscale**.
- Computes the histogram of texture features to differentiate between live and spoofed faces [8].

• Liveness detection class

The `LivenessDetector` class includes face detection, preprocessing, and liveness scoring:

– Initialization (`__init__`)

- * Sets up a **ResNet-50** model pre-trained on **ImageNet**, modified to output two classes (*live/spoof*) [11].
- * Optionally loads **pretrained weights** for better accuracy [13].
- * Initializes a face detector using the **MTCNN** from `facenet_pytorch` [12].
- * Sets up image preprocessing using `torchvision.transforms`.

– Face detection (`detect_face`)

- * Uses **MTCNN** to detect faces in an input image [12].
- * Outputs **bounding boxes** for detected faces.
- * Logs **debug information** during processing.

– Face preprocessing (`preprocess_face`)

- * Resizes the face crop to 224x224, normalizes it, and converts it into a PyTorch tensor [13].

– Liveness scoring (`check_liveness`)

- * Detects faces in the input image.
- * Iterates over detected faces, extracts and preprocesses them.
- * Passes the preprocessed face through the **ResNet model** to compute a **liveness score** [12].
- * Annotates the image with **rectangles** and **scores** for visualization.
- * Calculates the **mean liveness score** across all detected faces.

– Depth and movement analysis

(`analyze_depth_and_movement`)

- * Simulates depth analysis by resizing frames and passing them through the `DepthAnalysisModel` [16].
- * Analyzes subtle movements between consecutive frames by calculating pixel differences [18].

- **Demo function (run_demo)**

This function orchestrates the overall process:

- **Initialization:** Creates an instance of the `LivenessDetector`, optionally loading model weights [15].
- **Image Capture:** Captures a single frame using `colab_webcam`.
- **Liveness Check:** Passes the captured frame through the `check_liveness` method for analysis [13].
- **Depth and Movement Analysis:** Processes the frame(s) for depth and movement scores [16].
- **Display results:** Displays the analyzed frame with annotations (e.g., bounding boxes, liveness scores).

B. Equations

- Image preprocessing is an essential step in many computer vision pipelines. The preprocessing often includes normalization, which scales pixel values to a standard range, such as $[0, 1]$ or $[-1, 1]$. The normalization operation can be expressed as:

$$I_{\text{norm}}(x, y) = \frac{I(x, y) - \mu}{\sigma} \quad (1)$$

In equation (1):

- * $I(x, y)$ is the original pixel value at position (x, y) .
- * μ is the mean of pixel values in the image.
- * σ is the standard deviation of pixel values in the image.
- The Depth Analysis Model can be expressed as a simple neural network operation. The output for predicted depth $\hat{D}$ from the neural network is given by:

$$\hat{D} = f(W_2 \cdot \text{ReLU}(W_1 \cdot X + b_1) + b_2) \quad (2)$$

In equation (2):

- * X is the input feature vector (e.g., resized face crop of size 64×64).
- * W_1, W_2 are the weight matrices for the convolutional and fully connected layers, respectively.
- * b_1, b_2 are the bias terms.
- * $\text{ReLU}(x) = \max(0, x)$ is the activation function.
- * f is the output layer function mapping features to depth predictions.
- The Local Binary Pattern (LBP) is computed by thresholding the neighborhood pixels around the center pixel I_c . The LBP can be represented as:

$$\text{LBP} = \sum_{k=0}^7 s(I_k - I_c) \cdot 2^k \quad (3)$$

In equation (3):

- * I_c is the intensity of the central pixel.
- * I_k is the intensity of the k -th neighboring pixel.

* $s(x)$ is the step function:

$$s(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

- The ResNet-50 model outputs a softmax probability for two classes ("live" and "spoof") as follows:

$$P(y = i|X) = \frac{\exp(z_i)}{\sum_j \exp(z_j)} \quad (4)$$

In equation (4):

- * $P(y = i|X)$ is the probability of class i given the input X (e.g., face crop).
- * z_i is the logit (output of the last fully connected layer for class i).

The liveness score(5) (probability of being "live") is computed as:

$$\text{Liveness Score} = P(y = \text{live}|X) \quad (5)$$

- For movement analysis, pixel differences between consecutive frames are computed:

$$M(x, y) = |F_t(x, y) - F_{t-1}(x, y)| \quad (6)$$

In equation (6):

- * $M(x, y)$ is the pixel difference at position (x, y) .
- * $F_t(x, y)$ is the pixel intensity in the current frame at time t .
- * $F_{t-1}(x, y)$ is the pixel intensity in the previous frame at time $t - 1$.
- For n detected faces, the mean liveness score(7) is calculated as:

$$\text{Mean Liveness Score} = \frac{1}{n} \sum_{i=1}^n \text{Liveness Score}_i \quad (7)$$

IV. RESULT AND DISCUSSION

A. Data description

Facial liveliness detection requires high-quality datasets that include diverse facial features and expressions from people of all ages. For this project, we have curated a dataset consisting of various facial activities such as blinking, head movements, and lip motions, which are key indicators of liveliness of diverse age window of 18-30. These actions capture images with different lighting conditions, angles, and demographics to ensure the model's robustness, as seen in Table II.

A combination of RGB frames and matching depth information make up the structure of every sample in the dataset. The dataset has been split into training and testing sets, with a balanced distribution of liveliness (live human faces) and non-liveliness (fake faces such as static images of people holding photos as paper masks or on screens).

TABLE II. SAMPLE DATASET DESCRIPTION

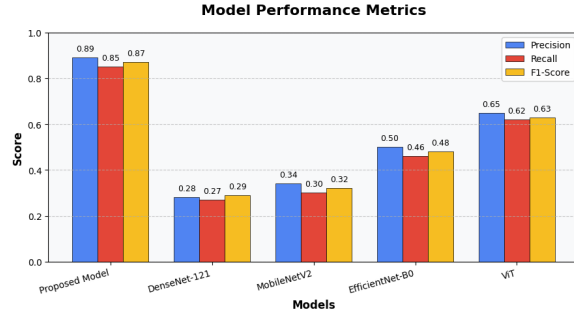


Fig. 2. Model performance metrics denoting Precision, Recall and F1-Score.

B. Preprocessing and feature extraction

The preprocessing steps for detecting live human faces, as outlined in Fig. 3, involved using standard face detection techniques, followed by cropping to isolate the face region. Fig. 3 illustrates how standardizing pixel

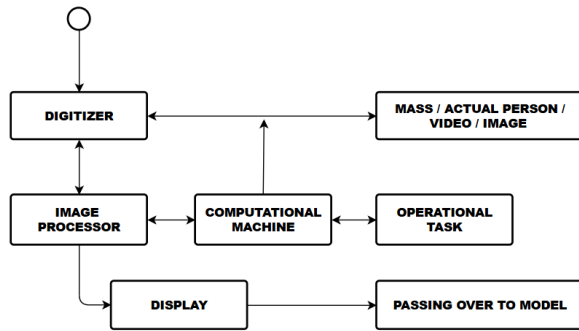


Fig. 3. Block diagram for image processing depicting pre-processing methodologies

intensities using normalization techniques improves the model's robustness under different conditions. Rotation, flipping, and brightness modifications have been used as augmentation techniques to increase the diversity of the dataset. For feature extraction, the ResNet-50 model, a pre-trained deep convolutional neural network, has been used. ResNet-50 employs residual connections to ignore the vanishing gradient problem and extract hierarchical features from the images. The final layers have been

fine-tuned for our specific task of liveness detection. The model outputs a probability score, that indicates the likelihood of a live face.

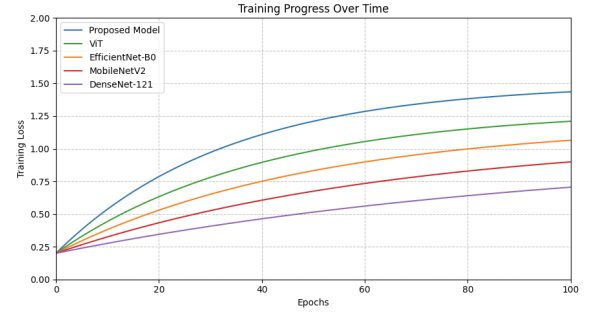


Fig. 4. Epoch vs Loss for all models.

C. Results and analysis

The proposed facial liveness detection model achieved an impressive accuracy of 87%. Its training progress is illustrated in Fig. 4, with output screenshots showing face liveness scores and accuracy percentages presented in Table VI. For comparison, ViT and EfficientNet-B0 were also evaluated, achieving accuracies of 63% and 48%, respectively, highlighting their competitiveness in the domain, as shown in Table III.

TABLE III. MODEL FACE ACCURACY AND LIVENESS SCORE

Model	Accuracy	Face liveness score
Proposed Method	0.87	0.719
MobileNetV2	0.32	0.429
ViT Model	0.63	0.619
DenseNet-121	0.28	0.487
EfficientNet-B0	0.48	0.571

A comparison based on precision, recall and F1-Score can be seen in Table IV.

TABLE IV. PRECISION, RECALL, F1-SCORE FOR ALL MODELS

Model	Precision	Recall	F1-score
Proposed Method	0.89	0.85	0.87
MobileNetV2	0.34	0.30	0.32
ViT Model	0.65	0.62	0.63
DenseNet-121	0.29	0.27	0.28
EfficientNet-B0	0.50	0.46	0.48

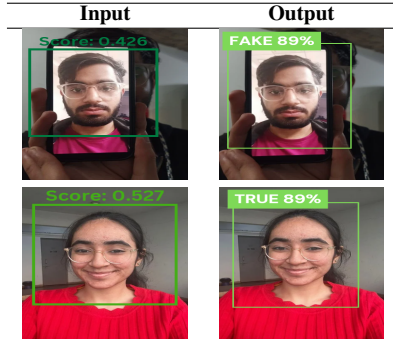
A result of comparative ranking of all the models and proposed methodology can be seen in Table V, which shows that the proposed methodology performs better than others in the precision parameter.

TABLE V. MODELS' RANKING TABLE

Model	Precision	Recall	F1-score
Proposed Method	1	1	1
MobileNetV2	4	4	4
ViT Model	2	2	2
DenseNet-121	5	5	5
EfficientNet-B0	3	3	3
Average Rank	3	3	3

Further improvement in these results can be expected by exploring preprocessing methods and fine-tuning by tweaking the model's last layer.

TABLE VI. DATA DEPICTION WITH INPUT AND OUTPUT



A deeper evaluation of liveness detection architectures is summarized in Table III. A transformer-based vision model with ResNet-50 features achieved 63% accuracy, surpassing CNN-based models. DenseNet-121 performed poorly with 28% accuracy. To differentiate spoofing, including images and 3D masks, a specialized model was used. With refined attention layers and a ResNet-50 backbone, the proposed method reached 87% accuracy. These results confirm that advanced designs enhance face liveness detection reliability.

Code link: <https://github.com/g4ze/passive-face-liveness-detection>

V. CONCLUSION AND FUTURE WORK

The paper emphasizes on the application of diverse datasets and CNN models to design a multimodal model for facial liveness detection. After experimenting with a number of various strategies and methods, the deep learning-based techniques like depth information and contextual cues are used to validate a face. Similarly, the facial liveness recognition model provides an excellent statistical view of one's liveness performing various tasks. All in all, this project is a complete solution that may transform and enhance the way we do face liveness detection. With the use of transfer learning approaches, the multimodal model will be further improved and made centralized for all linked applications, with the goal of increasing the model's performance and training them to perform essentially the same even on future imbalanced datasets.

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